

Statistical Analysis of T-zone Enrichment in the Gillespie Simulation

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Abstract

This document summarizes the statistical analysis of the observed cell-death locations generated by a Gillespie-type simulation with caspase activation, cell death, and ERK negative feedback. The main question is whether the observed death process is enriched in a fixed T-shaped spatial region. The analysis is based on a burn-in period of 500 observed deaths, followed by 1000 analyzed observed deaths for each replicate. For each parameter setting, five independent replicates were simulated and then pooled. The number of deaths falling inside the T-zone was compared with the geometric null probability using a one-sided exact binomial test. Since 52 parameter settings were tested, Bonferroni correction was applied to control the global significance level. The results show that the baseline setting does not produce a robust T-zone enrichment. In contrast, very strong and statistically significant T-zone enrichment appears when activation centers are sufficiently localized in space and/or short-lived in time.

1 Simulation and Statistical Objective

The simulation is performed on the spatial domain

$$W = [0, 10] \times [0, 10],$$

so that

$$|W| = 100.$$

A fixed T-shaped region is used as the background region where activation can be stronger. This region is denoted by T . In the present geometry, the T-zone consists of the middle vertical column of a 3×3 grid, together with the left middle arm. Hence

$$|T| = \frac{10}{3} \times 10 + \frac{10}{3} \times \frac{10}{3} = \frac{100}{3} + \frac{100}{9} = \frac{400}{9}.$$

Therefore the area fraction of the T-zone is

$$p_0 = \frac{|T|}{|W|} = \frac{400/9}{100} = \frac{4}{9} \simeq 0.4444.$$

For each simulation run, the first 500 observed deaths are discarded as a burn-in period. The next 1000 observed deaths are used for the statistical analysis. Therefore, for each replicate,

$$n = 1000.$$

For each parameter setting, five independent replicates are available. In the pooled analysis, this gives

$$N = 5 \times 1000 = 5000.$$

The central statistical question is whether the observed deaths are more concentrated in the T-zone than expected under a spatially uniform null model.

2 T-zone Enrichment Statistic

Let

$$K = \#\{\text{observed deaths in the T-zone}\}$$

be the number of analyzed observed deaths located inside the T-zone. Under the null hypothesis of spatial uniformity with respect to the T-zone indicator, we use

$$K \sim \text{Binomial}(n, p_0),$$

where

$$p_0 = \frac{|T|}{|W|}.$$

The hypotheses are

$$H_0 : \pi = p_0, \quad H_A : \pi > p_0,$$

where π denotes the true probability that an observed death falls inside the T-zone.

For one replicate, the one-sided exact binomial p-value is

$$p_{\text{val}} = \mathbb{P}_{H_0}(X \geq K), \quad X \sim \text{Binomial}(n, p_0).$$

For the pooled analysis over five replicates, the same formula is used with

$$n = N = 5000.$$

In addition to the binomial p-value, we also compute the T-zone density ratio

$$\rho_T = \frac{K/|T|}{(n - K)/(|W| - |T|)}.$$

Equivalently,

$$\rho_T = \frac{K/p_0}{(n - K)/(1 - p_0)}.$$

A value

$$\rho_T > 1$$

means that observed deaths are denser in the T-zone than outside the T-zone. A value close to 1 means that the T-zone has approximately the same death density as the rest of the domain.

3 Multiple Testing Correction

Since the parameter sweep contains 52 parameter settings, the pooled binomial test is repeated 52 times. To avoid interpreting random false positives as true discoveries, Bonferroni correction is applied. With a global significance level

$$\alpha = 0.05,$$

the Bonferroni threshold is

$$\alpha_{\text{Bonferroni}} = \frac{0.05}{52} = 0.000961538.$$

A parameter setting is therefore considered statistically significant only if

$$p_{\text{pooled}} < 0.000961538.$$

4 Baseline and Main Control Scenarios

The baseline setting does not show robust T-zone enrichment. Although one single baseline replicate gives a nominally significant p-value, this effect does not remain stable across the five independent replicates. In the pooled analysis, the baseline result is

$$K = 2187, \quad N = 5000,$$

so that

$$\hat{\pi}_T = \frac{2187}{5000} = 0.4374.$$

This is slightly below the geometric null probability

$$p_0 = 0.4444.$$

The pooled one-sided binomial p-value is

$$p_{\text{pooled}} = 0.845,$$

which is far above the Bonferroni threshold. Therefore, the baseline case does not provide evidence of T-zone enrichment.

The same conclusion holds for several simple parameter modifications. The higher death-rate scenario has pooled fraction

$$\hat{\pi}_T = 0.4438,$$

with pooled p-value

$$p_{\text{pooled}} = 0.542.$$

This is close to the null area fraction and is not significant. Thus increasing λ_d mainly accelerates the temporal occurrence of deaths, but it does not create a robust spatial enrichment in the T-zone.

Similarly, the larger ERK radius and shorter ERK duration scenarios are not significant after correction. The larger ERK radius scenario gives

$$\hat{\pi}_T = 0.4168, \quad p_{\text{pooled}} \simeq 1.$$

The shorter ERK duration scenario gives

$$\hat{\pi}_T = 0.4410, \quad p_{\text{pooled}} = 0.693.$$

These results suggest that ERK parameters have a strong effect on the temporal rate of observed deaths, but do not by themselves generate a robust T-zone spatial enrichment.

Parameter setting	Pooled K/N	Fraction	Pooled p-value	Significant?
Baseline	2187/5000	0.4374	0.845	No
Higher death rate	2219/5000	0.4438	0.542	No
Larger ERK radius	2084/5000	0.4168	1.000	No
Shorter ERK duration	2205/5000	0.4410	0.693	No
Stronger T-zone activation	2107/5000	0.4214	1.000	No

Table 1: Pooled results for the baseline and main control scenarios. None of these settings is significant after Bonferroni correction.

5 Effect of Increasing the T-zone/Outside Activation Contrast

A natural first hypothesis is that increasing the ratio

$$\frac{\lambda_a^T}{\lambda_a^c}$$

should automatically create T-zone enrichment. However, the results show that this is not sufficient when the activation zones remain large or long-lived.

For example, in the setting

$$\lambda_a^T = 0.5, \quad \lambda_a^c = 0.005,$$

the ratio is

$$\frac{\lambda_a^T}{\lambda_a^c} = 100.$$

Nevertheless, the pooled T-zone fraction is only

$$\hat{\pi}_T = 0.4420,$$

with pooled p-value

$$p_{\text{pooled}} = 0.641.$$

This is not significant.

Even when the ratio is increased to 1000 or 10000, the pooled fractions remain below or close to the null area fraction:

$$0.4238, \quad 0.4238.$$

Thus, increasing the activation contrast alone is not enough to produce a detectable T-zone death enrichment.

This indicates that the spatial structure of the active zones matters. If active disks are too large or persist for too long, the local T-zone signal is spatially smeared out. As a result, the death process does not remain strongly localized inside the fixed T-shaped region.

6 Strong T-zone Enrichment Under Local Activation

A very different behavior appears when the activation process is made more local. In the local-activation scenarios, the activation radius is reduced and the activation lifetime is shortened. These scenarios give extremely strong T-zone enrichment.

For example, the setting

$$T0p5_c0p005_ratio_1e2_local_activation$$

has

$$K = 4525, \quad N = 5000,$$

so that

$$\hat{\pi}_T = 0.9050.$$

This is much larger than the null probability $p_0 = 0.4444$. The pooled p-value is numerically reported as zero, meaning that it is below machine precision in the computation. The pooled density ratio is approximately

$$\rho_T \simeq 11.91.$$

Similarly,

$$T0p5_c0p0005_ratio_1e3_local_activation$$

gives

$$K = 4548, \quad N = 5000, \quad \hat{\pi}_T = 0.9096, \quad \rho_T \simeq 12.58.$$

The most extreme local-activation scenario considered here,

$$T1e-1_cle-4_ratio_1e3_local_activation,$$

gives

$$K = 4661, \quad N = 5000, \quad \hat{\pi}_T = 0.9322, \quad \rho_T \simeq 17.19.$$

All these results remain significant after Bonferroni correction.

Parameter setting	Pooled K/N	Fraction	Density ratio	Significant?
Strong visible T-zone	4525/5000	0.9050	11.91	Yes
Local activation, ratio 10^2	4525/5000	0.9050	11.91	Yes
Local activation, ratio 10^3	4548/5000	0.9096	12.58	Yes
Low absolute intensity, local activation	4661/5000	0.9322	17.19	Yes

Table 2: Pooled results for the local-activation scenarios. These settings show very strong T-zone enrichment and remain significant after Bonferroni correction.

7 Effect of the Activation Radius and Activation Lifetime

The most informative part of the parameter sweep is the scan over β_a^R and β_a^T . The activation radius satisfies

$$R^a \sim \text{Exp}(\beta_a^R), \quad \mathbb{E}[R^a] = \frac{1}{\beta_a^R}.$$

Therefore, increasing β_a^R decreases the mean activation radius and makes activation more spatially localized.

The activation lifetime is controlled by β_a^T , with

$$\mathbb{E}[T^a] = \frac{1}{\beta_a^T}.$$

Thus, increasing β_a^T shortens the lifetime of active centers.

The results show a clear localization effect. When β_a^R and β_a^T are small, the T-zone density ratio is close to one and there is little or no robust enrichment. When β_a^R and/or β_a^T increase, the T-zone density ratio grows rapidly.

For the family

$$\lambda_a^T = 0.5, \quad \lambda_a^c = 0.005,$$

the mean T-zone density ratio behaves as follows:

$\beta_a^R \backslash \beta_a^T$	1.2	2.0	3.0	5.0
2.5	0.993	1.190	2.164	4.202
3.5	1.285	4.296	6.236	8.146
5.0	8.349	11.304	11.930	13.226
7.5	18.376	17.122	22.999	22.149

This table shows that the T-zone signal becomes much stronger when the activation radius is smaller. The increase in β_a^R has a particularly strong effect. Increasing β_a^T , which shortens activation lifetimes, also tends to increase the T-zone signal, especially when the activation radius is not yet very small.

A similar pattern appears in the lower absolute activation-intensity family

$$\lambda_a^T = 0.1, \quad \lambda_a^c = 10^{-4}.$$

The mean T-zone density ratios are

$\beta_a^R \backslash \beta_a^T$	1.2	2.0	3.0	5.0
2.5	0.939	1.054	2.359	4.492
3.5	1.300	5.325	7.666	9.858
5.0	9.477	11.927	17.692	15.517
7.5	21.977	21.315	23.462	25.144

Again, the strongest enrichment appears when β_a^R is large, i.e. when the activation radius is small.

8 Interpretation

The statistical results support the following mechanism.

First, the baseline simulation does not produce stable T-zone enrichment. This means that the fixed T-shaped background activation zone is not automatically visible in the observed death process. A single significant replicate should not be over-interpreted, because the pooled result over five independent replicates is not significant.

Second, increasing the T-zone activation intensity alone is not sufficient. Even large values of

$$\lambda_a^T / \lambda_a^c$$

do not necessarily produce T-zone enrichment if active zones are too large or too persistent. In that case, the activation field spreads beyond the T-zone, and the spatial signal is diluted.

Third, the T-zone enrichment becomes very strong when activation centers are localized. This happens when the mean activation radius $1/\beta_a^R$ is small and/or when the mean activation lifetime $1/\beta_a^T$ is short. Under these conditions, the death process remains closer to the spatial structure of the fixed T-zone, and the T-zone enrichment becomes statistically obvious.

Therefore, the main conclusion is not simply that the T-zone activation intensity is higher. The more precise conclusion is that the T-zone signal is visible only when the activation process is sufficiently local in space and time.

9 Conclusion

The pooled Bonferroni analysis leads to three main conclusions.

First, the baseline parameter setting does not produce robust T-zone enrichment. The pooled baseline fraction is

$$0.4374,$$

which is slightly below the null area fraction

$$p_0 = 0.4444,$$

and the pooled p-value is

$$0.845.$$

Thus, the baseline result is compatible with the spatially uniform null model with respect to the T-zone indicator.

Second, changing temporal or ERK-related parameters mainly changes the speed of the observed death process, not the spatial T-zone enrichment. Increasing λ_d makes deaths occur

faster, and changing ERK radius or duration affects the rejection and protection dynamics, but these changes do not create a robust T-zone signal by themselves.

Third, strong and statistically significant T-zone enrichment appears when the activation process becomes spatially localized. The clearest evidence comes from the local-activation scenarios and from the (β_a^R, β_a^T) sweep. Increasing β_a^R , which decreases the mean activation radius, is especially important. Increasing β_a^T , which shortens the activation lifetime, also contributes to making the T-zone signal clearer.

Overall, the analysis shows that T-zone enrichment is not merely a consequence of a larger activation intensity in the T-zone. It is a consequence of the interaction between the fixed T-shaped activation landscape and the spatial locality of the activation centers. The observed death process becomes strongly enriched in the T-zone only when activation disks are sufficiently small or short-lived.